

CS & IT ENGINEERING



Operating System

Process Management

Lecture -2

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Recap of Previous Lecture



Doubt ?



Topic

Dual Mode of Operation

Topic

Process

Topic

Process Representation

Topic

Process Control Block

Topics to be Covered



Topic

Process states

Topic

Process State Transition

Topic

Process Scheduling

Topic

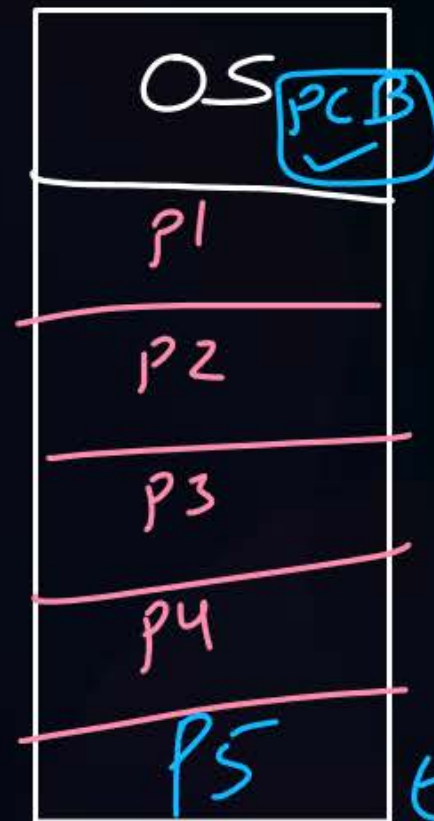
Types of Schedulers



Topic : Process States



defined based on what process is doing now.



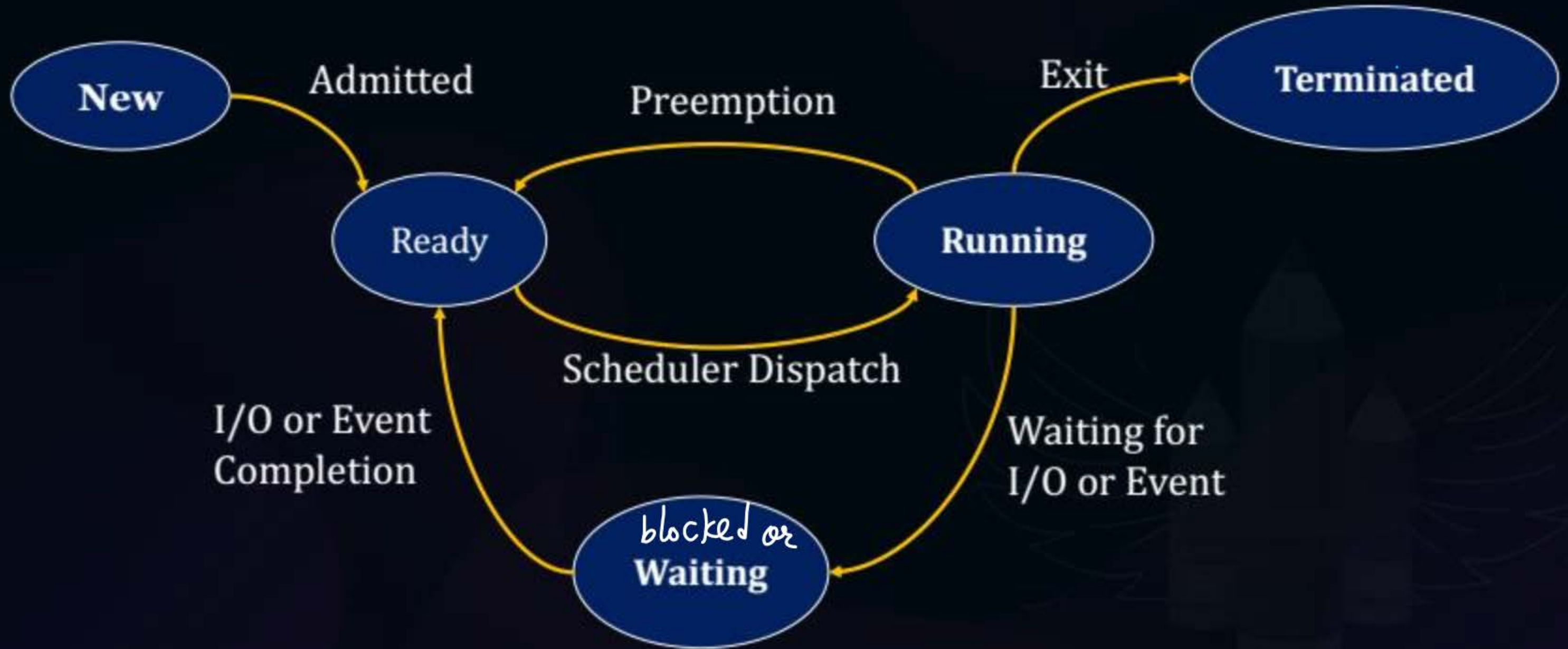
Hard disk





Topic : Process States

Preemptive system





Topic : Process States

New:- All installed processes are known to be in new state (not in execution)

Ready:- All processes which are waiting to run on CPU are known to be in ready state

Running:- A process which is running on CPU has its state as running

Terminated:- A completed process has its state as terminated

Blocked:- All processes which are waiting for any I/O or event



Topic : Process States



New To Ready: When process is admitted by OS $\Rightarrow$ mem. and other resource allocation

Ready to Running: When a process is dispatched to CPU

Running to Terminated: When a process is completed $\Rightarrow$ mem. and all other resources are deallocated

Running to Blocked: When a process goes for IO or event

Running to Ready: When a process is preempted

Blocked to Ready: When a process completes IO or event



Topic : Process States



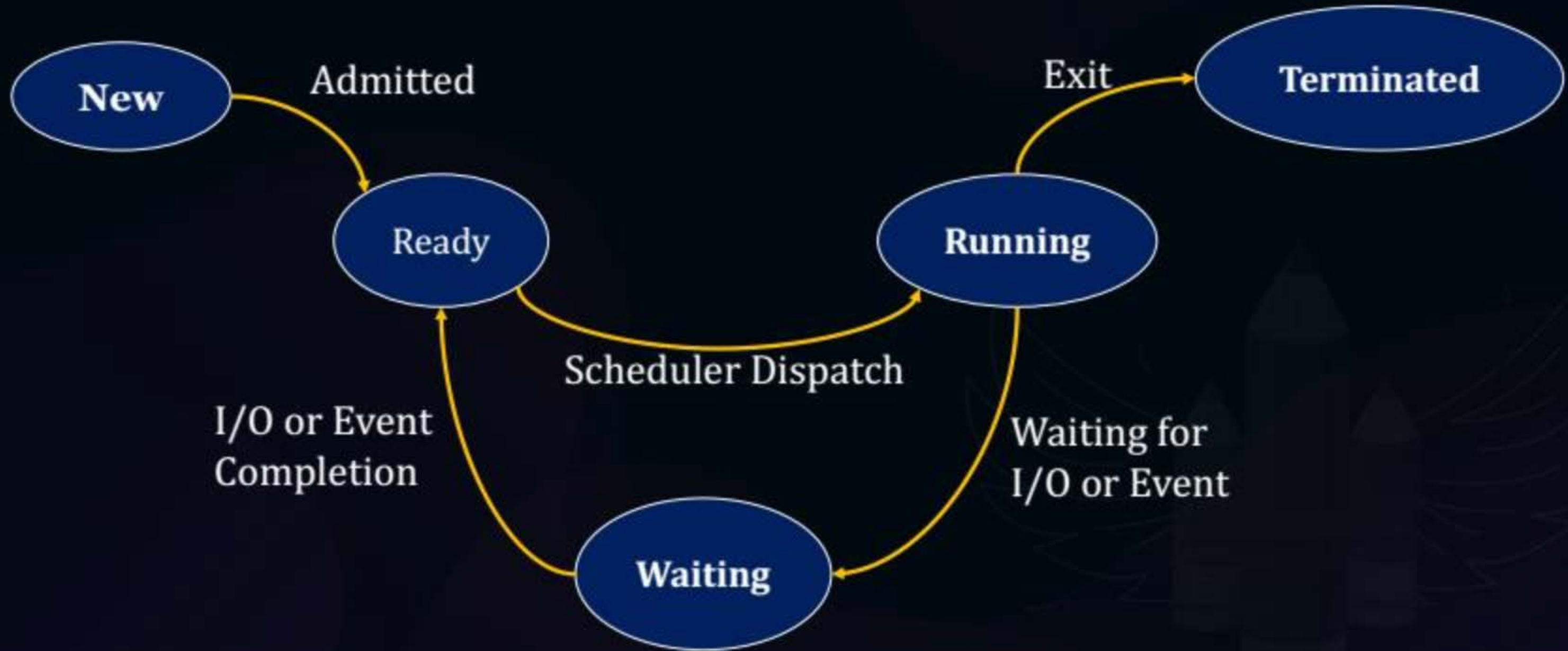
- ❑ 2 Transitions are voluntary:
 - Running to Terminated
 - Running to Blocked

Process will be in mm when in

→ Ready state
→ Running state
→ Blocked state



Topic : Non-preemptive Process States





Topic : CPU vs IO Bound Process

CPU Bound: If the process is intensive in terms of CPU operations

IO Bound: If the process is intensive in terms of IO operations





Topic : Process Scheduling



Needed Because?



to extract better efficiency and outcome.



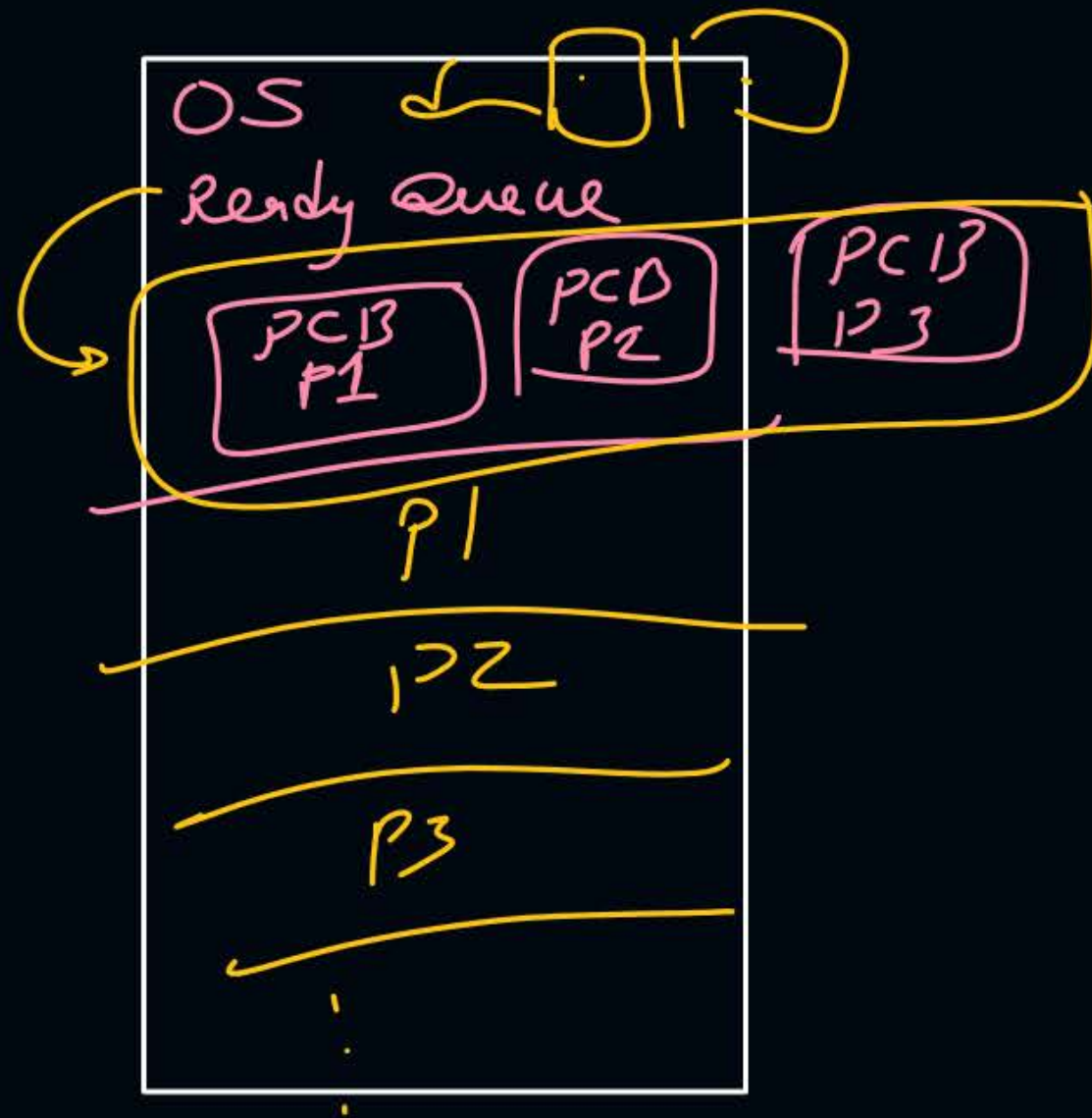


Topic : Scheduling Queues

- only 1
- Job Queue $\Rightarrow$ list where all the new state processes^{names} are kept.
 - Ready Queue $\Rightarrow$ list where all the ready state ——— || ——— (Process's PCB)
 - Device Queue $\Rightarrow$ list where all the blocked processes are kept which are waiting for a device.

↓
one for each device.

ex:- P_1, P_2, P_3 are in ready state





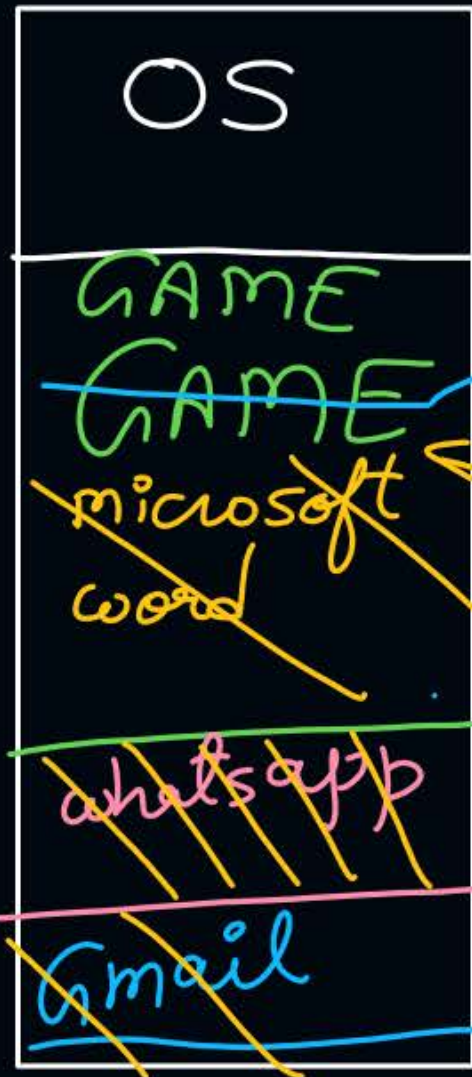
Topic : Types of Schedulers

- Long-Term Scheduler (Job) → schedules one of new processes to ready state
- Short-Term Scheduler (CPU) → selects one of ready processes to run on CPU
- Mid-Term Scheduler (Medium-term) → it performs swapping (swap-out and swap-in)

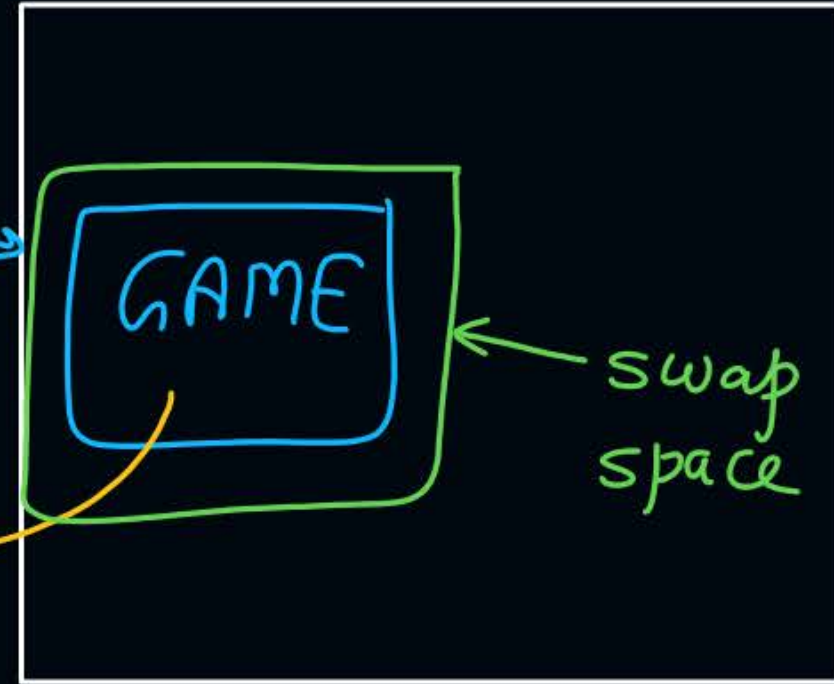
⇒ Long-term scheduler controls degree of multiprogramming.

⇒ Dispatcher brings the process's context inside CPU to run.

RAM (memory)



Disk



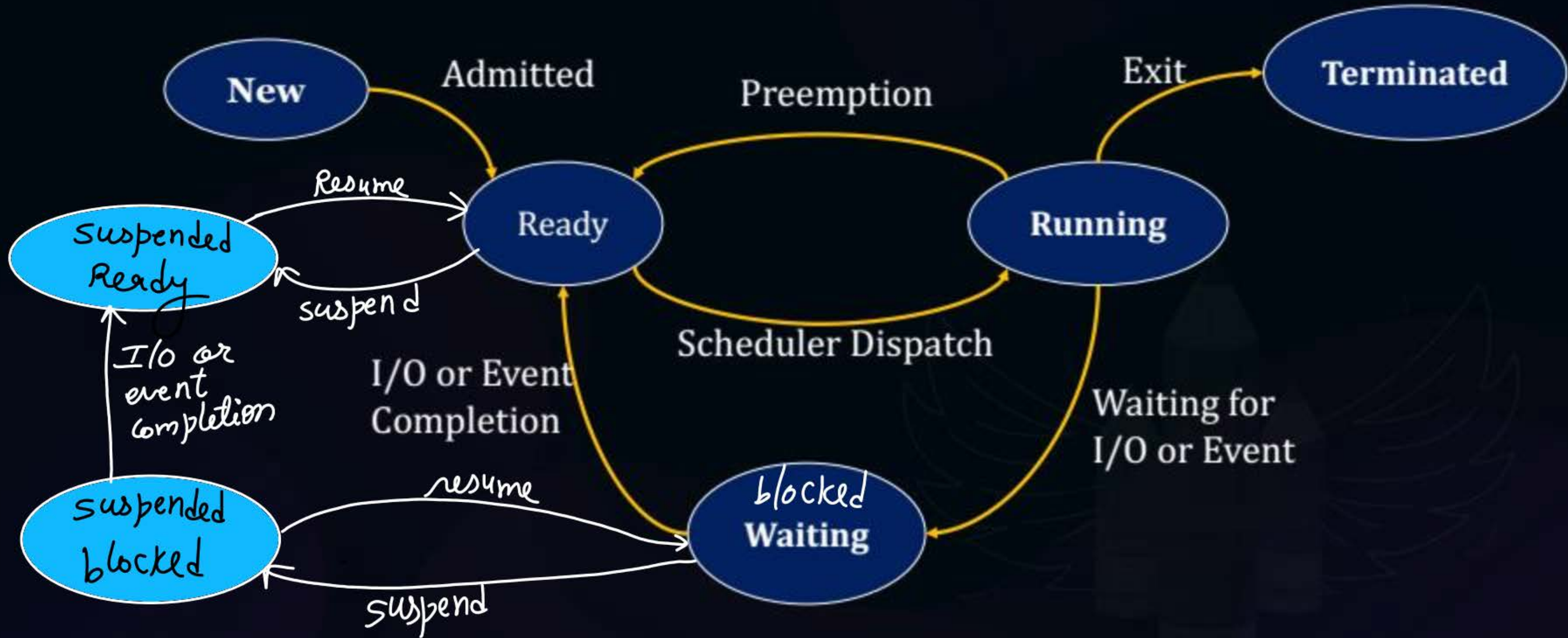
swap out

swap in



Topic : Process States

updated



[MCQ]

#Q. Which of the following scheduler^{Can} reduce the degree of multiprogramming?

- A** Short-Term $\Rightarrow$ does not change
- B** Long-Term $\Rightarrow$ Can only increase
- C** ☒ Mid-Term $\Rightarrow$ Can increase also and can decrease also
- D** Long-Term and Mid-Term both



2 mins Summary

Topic

Process states

Topic

Process State Transition

Topic

Process Scheduling

Topic

Types of Schedulers



Happy Learning

THANK - YOU

